COURSE TITLE : MECHANICAL & ELECTRICAL ENGINEERING
COURSE CODE : 3121
COURSE CATEGORY : B
PERIODS PER WEEK : 4
SEMESTER : 3
PERIODS PER SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Fluid Mechanics , Pumps, Fans, Compressor Transmission of Power Test	14 1
2	Properties of steam Steam Generators Boiler mounting and accessories Test	14 1
3	**ELECTRICAL ENGINEERING** Basic Principles and DC circuits AC Fundamentals Test	14 1
4	Electrical Machines, Measuring Instruments and electronics Test	14 1
	Total	**60**

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand the basic principles of fluid mechanics. principle of operations of pumps, fans and compressors.
	2	Understand Pumps and Compressors
2	1	Understand the various properties of steam.
	2	Understand the working of steam generators.
3	1	Understand Basic principles of D.C Circuits and A C Fundamentals.
	2	Understand Work, Power and Energy

4	1	Understand the electrical machines, Electrical measuring Instruments and Basic electronics.
	2	Understand DC Motors
	3	Understand Electrical Measuring Instruments

Specific outcome

PART I:- Mechanical Engineering

MODULE I

1.1.0 Understand the basic principles of fluid mechanics. principle of operations of pumps, fans and compressors.

- **1.1.1** Define fluid, Newtonian and non-Newtonian fluids, density, specific weight, specific gravity, specific volume, compressibility, surface tension, capilarity and viscosity, Kinematic viscosity, intensity of pressure and discharge of liquid.
- **1.1.2** Determine the fluid pressure using piezometer and simple U-tube manometer.
- **1.1.3** Determine the difference of pressure using U-tube differential manometer.
- **1.1.4** State the equation of continuity of flow.
- **1.1.5** Define laminar flow, turbulent flow, critical velocity and Reynold's number.
- **1.1.6** State Potential energy, kinetic energy, pressure energy Bernoulli's equation.
- **1.1.7** Solve simple problems using Bernoulli's equation.
- **1.1.8** State the working principle if venturimeter, small orifice and pitot tube. State Darcy's formula compute frictional loss in pipes.

1.2.0 Understand Pumps and Compressors

- **1.2.1** Classify pumps, fans, and compressors.
- **1.2.2** Explain working of single acting reciprocating pump, double acting reciprocating pump, gear pump, vane pump, screw pump, centrifugal pump, single stage and multistage centrifugal pump, single stage and multistage centrifugal pumps, centrifugal fan, axial flow fan, blowers, reciprocating compressor, rotary compressor, vaccum pump and ejectors.
- **1.2.3** Distinguish between open impeller and closed impeller in a centrifugal pump
- **1.2.4** Describe the uses of power transmission devices.
- **1.2.5** Explain the applications of belt and chain drives.
- **1.2.6** List the various types of belt drives.
- **1.2.7** Explain the terms open belt, cross belt, length of belt Vee belt, slack and tight side, Ratio of tension Velocity ratio.
- **1.2.8** Compute the power transmitted in belt drive, length of belt and thickness of belt.
- **1.2.9** Explain the different types of chains such as Roller chain and silent chain.
- **1.2.10** Calculate velocity ratio and power transmitted
- **1.2.11** Describe the advantages and disadvantages of gear drive

1.2.12 Explain simple gear train, compound gear train, reverted gear train and epycyclic gear train

MODULE II

2.1.0 **Understand the various properties of steam.**

2.1.1 Explain the formation of steam under constant pressure and constant temperature.
2.1.2 Describe the effect of pressure and temperature
2.1.3 Distinguish the wet steam, dry steam and super heated steam.
2.1.4 Explain the properties of steam such as total heat of water, latent heat, Total heat of steam
(wet, dry and super heated)
2.1.5 State the dryness function of steam.
2.1.6 Compute the total heat of steam using steam tables.

2.2.0 **Understand the working of steam generators.**

2.2.1 Define a boiler
2.2.2 Explain the functions of a boiler
2.2.3 Distinguish between the different types of boilers.
2.2.4 Differentiate the fire tube boiler and water tube boiler.
2.2.5 Describe with a line diagram the construction and working of a Cochran boiler.
2.2.6 Explain with a line diagram the construction and working of a Babcock and Wilcox boiler.
2.2.7 Explain the working of La – Mont. boiler with the aid of a line diagram.
2.2.8 Understand the functions of boiler mountings and accessories.
2.2.9 Describe the functions of boiler mountings such as stop valve, safety valve, water level indicator and pressure gauge with line sketches.
2.2.10 Describe the functions of boiler accessories such as economiser, feed pump, and super heater with simple sketches.
2.2.11 Compare the boiler mounting and accessories.

Part:2 Electrical Engineering

MODULE III

3.1.0 **Understand Basic principles of D.C Circuits and A C Fundamentals.**

3.1.1 Distinguish between emf and pd.
3.1.2 State e.m.f current and resistance
3.1.3 State the laws of resistance
3.1.4 State Ohm's law as applied to electric circuits
3.1.5 Compute the effective resistance of a given series and parallel D.C circuits.
3.1.6 State the Faraday's law of electromagnetic induction.
3.1.7 Distinguish between self and mutual inductance.

3.2.0 **Understand Work, Power and Energy**

- 3.2.1 State work, power and energy.
- 3.2.2 Solve simple problem using work power and energy.
- 3.2.3 Describe the generation of sinusoidal emf
- 3.2.4 Define cycle, frequency, time period, R M S, value, average value form factor.
- 3.2.5 State capacitance Compute the effective capacitance in series and parallel capacitor connective
- 3.2.6 Solve simple problems in RLC series circuits
- 3.2.7 State Solve simple problems
- 3.2.8 Draw the 3-phase power and state its working principle.
- 3.2.9 Differentiate phase and line value, voltage and current in star and delta connections.

MODULE IV

4.1.0 **Understand the electrical machines, Electrical measuring Instruments and Basic electronics.**

- 4.1.1 Draw the constructional details of transformers
- 4.1.2 Specify the various transformer
- 4.1.3 Derive the e.m.f equation.
- 4.1.4 State the various losses in the transformers
- 4.1.5 Compute the efficiency
- **4.1.6** State the working principle of autotransformers and its applications.

4.2.0 **Understand DC Motors**

- 4.2.1 Draw the constructional details of D C motors and name its parts.
- 4.2.2 State the working principle of 3-phase Induction motor.
- 4.2.3 Classify the types of induction motors.
- 4.2.4 Sketch DOL Starter, star Delta Starter, Auto-starter and their principles
- 4.2.5 Classify single-phase Induction motors.
- **4.2.6** Identify their field of application

4.3.0 **Understand Electrical Measuring Instruments**

- 4.3.1 Classify electrical instruments
- 4.3.2 Explain moving coil and moving iron ammeters and voltmeters with diagram
- 4.3.3 Explain dynamometer wattmeter and energy meter with diagrams.
- 4.3.4 List out various electron emission and compare them
- 4.3.5 Identify the P-type and N type materials
- 4.3.6 Draw the PN Junction diagram and state its operation
- 4.3.7 Draw V-I characteristic's of diode as forward and reverse biasing
- 4.3.8 Draw and explain half wave rectifier
- 4.3.9 Sketch and explain full wave rectifier

CONTENT DETAILS

Mechanical engineering

MODULE I

Fluid mechanics Fluid – classification Newtonian and non-Newtonian fluids – liquid and their properties density specific weight specific gravity, specific volume. Compressibility, surface tension, capillariry, viscosity, absolute viscosity and kinematic viscosity Fluid pressure and its measurements – Intensity of pressure, piezometer, simple u – tube manometer, differential u- tube manometer. Discharge of liquid – continuity equation. Types of flow – laminar and turbulent – critical velocity - Reynold's number Types of energy – Potential, kinetic and pressure energy. Bernaulli's equation (no proof) simple problems. Applications of Bernaulli's equations- venturimeter, orifice, pitot tube Flow through pipes – frictional loss in pipes – Darcy's formula (no derivation). Pumps, Fans and compressors Classification of pumps, positions displacement Centrifugal pump- reciprocating pump – single acting and double acting. Rotary pump – gear, vane, screw pump centrifugal pump principle of working priming of centrifugal pump. Types of impellers – open and closed. Single stage and multi stage centrifugal pumps fans – centrifugal axial flow. Blowers, compressors – reciprocating and rotary vacuum pumps – ejectors. Transmission of power. Belt and chain drives – explanation of the forms of open belt, cross belt, ver belt and tensioning belt – Calculation of power transmitted. Length of belt thickness of belt. Application of chain drives – Types of chain – Rollar and silent chain. Functions of gears – friction wheels – advantages and disadvantages of a gear drive – spur gear nomenclature – simple gear drive – velocity ratio – power transmitted – gear trains – simple gear train – compound gear train – reverted gear train – epicyclic gear train.

MODULE II

Steam Generators. Introduction - function – Classification – fire tube and water tube boilers – Cochran boiler, Bob cock and Wilcox boiler – modern high-pressure boiler – La – Mont boiler. Boiler, Mounting and accessories. Brief explanations and functions of boiler mountings – stop valve, Safety valve, water level indicator, Fusible plug, pressure gauge. Brief explanation and functions of boiler accessories – Feed pump, economiser, Super heater. Properties of steam. Introduction – Steam formation under constant temperature – wet steam, dry stam – dryness function – super heated steam. Properties – total heat of water, latent heat total heat of steam (wet,dry and super heated) use of steam tables – calculation of heat required for producing steam at a given pressure and state.

Electrical Engineering

MODULE III

Basic principles and dc circuits. Nature of electricity – concept of potential difference – current and voltage – definition and units. Resistance – laws of resistance – Ohm's Law – resistance in series and parallel – simple problems – Faraday's law of electromagnetic induction concept of self and mutual inductance units works and energy with simple problems AC Fundamentals Production of sinusoidal voltage, definition of cycle, frequency, time period RMS value, average, form factor etc. Capacitor – capacitance in series and parallel. Concept of impedance in R.L.C series circuits. Power and power factor. AC circuits – simple problems. Three phase system line and phase values of voltage and current in star and delta connected system

MODULE IV

Electrical Machines Transformers: - construction details – classification of transformers according to core fabrication EMF equation – losses, efficiency. Working principles of auto – transformer – fields of application. DC Motors – constructional details working principle. Types of DC motors series shunt component and fields of application of each type. Speed control of DC motor Three phase induction motors – working principles. Types and fields and application DOL starter, star delta starter. Autotransformer starting sketches. Principles of single phase induction motors - - types and method of starting, fields of application Measuring instruments and basic electronics Classification of electrical instruments – moving coil and moving iron ammeters and voltmeters. Dynamometer type watt meter (sketches and brief explanation only) Electronics: - electron emission – P – type and N – type material, PN junction – characteristics of diode. Forward and reverse biasing – applications, Half wave and full wave rectifier using diode

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4. V.K Metha.,Basic Electronics , S Chand
5. Sauhney, Electrical and Electronic Measurements
6. Khurmi, Theory of machines , S Chand
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8. B.L. Theraja, Basic electrical and electronics – S Chand
9. B.L. Theraja, Electrical technology Vol. I & II , S Chand
10. R.S.Khurmi, Fluid mechanics and machines S Chand ,
11. R.S.Khurmi, Thermal Engineering , S Chand
12. N.Narayana Pillai, Principles of Fluid Mechanics and Fluid Machin ,Universities Press